1. (7 points) What is computer security.

Computer security involves protecting computer systems and the data they store or process from unauthorized access, theft, damage, disruption, or manipulation. It includes various strategies, tools, and measures that ensure the confidentiality, integrity, and availability of information.

2. (6 points) List 3 types of computer security.

Security by Obscurity

Security by Legislation

Security by

3. (7 points) What is malware.

A generic term for software that are used for malicious reasonings.

4. (6 points) List 3 types of malware.

Spyware

Worms

Trajan Horse

5. (5 points) What is cyber security

Cybersecurity encompasses the practices, technologies, and processes designed to protect computers, networks, and data from cyber-attacks, unauthorized access, and digital threats. It involves ensuring the confidentiality, integrity, and availability of information.

6. (10 points) List 5 types of cyber-attacks.

Password cracking

Software cracking

Network cracking

Application cracking

Wireless cracking

7. (10 points) List 5 types of threats.

Errors of user

Natural/man-inside/machine disasters

Dishonest insider

Disgruntled insider

Outsiders

8. (3 points) Describe 3 type of attacks with examples

An interruption attack disrupts the availability or normal functioning of a system or service. For example, a Denial of Service (DoS) attack overwhelms a server with traffic, causing it to crash and become inaccessible to users.

Interception attacks involve unauthorized parties gaining access to data or communications. In a Man-in-the-Middle (MitM) attack, an attacker secretly intercepts and possibly alters the communication between two parties.

Modification attacks alter data without authorization, compromising its integrity. For example, a SQL Injection attack modifies a database query to manipulate or corrupt data, leading to unauthorized access or changes.

9. (6 points) List the three main security objectives.

Confidentiality

Integrity

Availability

10.(40 points) Explain how these security objectives are met by your personal computer. Be specific, e.g., availability of the system and data files is supported by regular system backups every week and by copying all data files to an online drive.

Confidentiality: On my personal computer, confidentiality is maintained through strong password protection, encryption of sensitive files, and using secure connections (HTTPS) when transmitting data online. Antivirus software scans files and communications to prevent unauthorized access.

Integrity: The integrity of my data is maintained by regularly updating software to patch vulnerabilities, using file integrity monitoring tools, and keeping backups that ensure data can be restored to a trustworthy state if altered.

Availability: Availability is ensured through regular system backups (e.g., weekly backups to an external hard drive and cloud storage), using a reliable power supply (like an uninterruptible power supply), and having redundant internet connections to avoid downtime during network failures.